

Topic 10: Ordinary Differential Equation¹

Differential equations are used in economics to model how variables evolve over time. They allow us to study dynamics such as economic and population growth, capital accumulation, the depletion of non-renewable resources, and many more. Before formally defining differential equations, we begin with a key concept: the growth rate.

GROWTH RATE

Let $x(t)$ represent the accumulated capital at time t . The change of capital over time is denoted as $\dot{x} = \frac{dx(t)}{dt}$.²

The **growth rate** is defined as $\frac{\dot{x}}{x}$. A useful formula for computing the growth rate is:

$$\frac{\dot{x}}{x} = \frac{d \log(x(t))}{dt}$$

We can verify this equation using the chain rule:

$$\frac{d \log(x(t))}{dt} = \frac{d \log(x)}{dx} \frac{dx(t)}{dt} = \frac{1}{x} \dot{x}$$

Thus, the derivative of $\log(x)$ with respect to time gives its growth rate.

ORDINARY DIFFERENTIAL EQUATIONS

Suppose we know that the capital grows at a constant rate ($a \in \mathbb{R}_{++}$):

$$\frac{\dot{x}}{x} = a$$

Can we find a function $x(t)$ that satisfies this equation? Yes. Using the equation above:

$$\frac{\dot{x}}{x} = \frac{d \log(x(t))}{dt} \Rightarrow \frac{d \log(x(t))}{dt} = a$$

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²This is the continuous-time analogue of $\Delta x_t = x_t - x_{t-1}$.

We integrate both sides and, by the Fundamental Theorem of Calculus:

$$\int \frac{d \log x(t)}{dt} dt = \int a dt \implies \log x(t) = at + C.$$

where C is some constant (called integration constant). Therefore, one solution to $\frac{\dot{x}}{x} = a$ is:

$$x(t) = e^{at+C}.$$

This function $x(t)$ satisfies the equation $\frac{\dot{x}}{x} = a$.

A **differential equation** is any equation that involves derivatives of a function. An **ordinary differential equation** (ODE) is an expression that describes the relationship between a single-variable function and its derivatives. A function that satisfies the equation is called a **solution** of the differential equation.

In the example of a capital constant growth rate, we solved the differential equation using direct integration. We will discuss some methods for solving differential equations, which will be especially useful for complex ones.

We now introduce some terminology commonly used in the study of differential equations.

Definition 1

Let $x(t)$ be a function differentiable with respect to t .

- An equation of the form

$$F(t, x) = \dot{x}$$

is called a first order **explicit ordinary differential equation**.

- If an equation can be written as

$$F(x) = \dot{x},$$

then we call it **autonomous** or **time-independent** differential equation.

- An equation of the form

$$F(t, x, \dots, x^{(n-1)}) = x^{(n)}$$

is called an **ordinary differential equation of order n** .

- If F is linear in x and its derivatives, the equation is called a **linear differential equation**.

Another important class in differential equations is the Partial Differential Equation (PDE), which involves a function of multiple independent variables and their partial derivatives. Here, we focus on ODE.

The following are some other examples of ordinary differential equations:

$$\dot{x} = tx, \quad \dot{x} = t^2, \quad \dot{x} = tx + t^2.$$

FIRST ORDER ODE

A first-order ordinary differential equation contains only the first derivative: $\dot{x}$. However, this derivative can enter in various powers: $\frac{dx(t)}{dt}$, $\left(\frac{dx(t)}{dt}\right)^2$, $\left(\frac{dx(t)}{dt}\right)^4$. The highest power attained by the derivative determines the *degree* of the differential equation. When the derivative $\frac{dx(t)}{dt}$ appears only in the first degree, and so does x , the equation is said to be *linear*.

The solutions of ODEs are sometimes far from trivial. Fortunately, for the linear ODEs, we have explicit solutions. In the following formulas, $a(t)$ and $b(t)$ are continuous functions.

Form 1 $\dot{x} = ax$.

It seems that we have found $x(t) = e^{at}$ as the solution. However, if you check more carefully,

$$x(t) = ke^{at}$$

for any $k \in \mathbb{R}$ can also be the solutions. Because we don't know the value of k , we call this a **general solution**.

To identify k , if we know the **initial condition** of $x(t)$, written as $x_0 = x(0)$, then we can solve k by

$$k = \frac{x(0)}{e^{a \cdot 0}} = x_0.$$

and we have:

$$x(t) = x_0 e^{at}$$

We call this a **particular solution**.

Example 1

Suppose the motion of capital, $\dot{k}$, satisfies the following equation: $\dot{k} = 0.03k$.

- (1) Find the general solution ($k(t)$) to this ODE.
- (2) Let $k(0) = 100$. Find the particular solution ($k(t)$) to this ODE.

Form 2 $\dot{x} = ax + b$.

The **general solution** is

$$x(t) = \frac{b}{a} + ke^{at}$$

for some k .

The **particular solution** (given initial condition $x(0) = x_0$) is:

$$x(t) = \frac{b}{a} + \left(x_0 + \frac{b}{a}\right)e^{at}.$$

Example 2

Suppose the motion of capital, $\dot{k}$, satisfies the following equation: $\dot{k} = 0.03k + 0.01$.

(1) Find the general solution ($k(t)$) to this ODE.

(2) Let $k(0) = 100$. Find the particular solution ($k(t)$) to this ODE.

Form 3 $\dot{x} = a(t)x$.

The **general solution** is

$$x(t) = ke^{\int^t a(s)ds}$$

for some k . The integral $\int^t a(s)ds$ is an indefinite integral with the upper limit t . According the Fundamental Theorem of Calculus, the derivative of the integral with respect to t is $a(t)$.

The **particular solution** (given initial condition $x(0) = x_0$) is:

$$x(t) = x_0 e^{\int^t a(s)ds}.$$

Form 4 $\dot{x} = a(t)x + b(t)$.

The **general solution** is

$$x(t) = \left[k + \int^t b(s)e^{-\int^s a(u)du}ds \right] e^{\int^t a(s)ds}$$

for some k .

For **Form 4**, we sometimes apply the **integrating factor** to solve it. As the name suggests, the method is to multiply some factor $r(t)$ to make

$$r(t)\dot{x} - r(t)a(t)x = r(t)b(t) \tag{1}$$

integrable. In fact, by product rule,

$$\frac{d}{dt}(r(t)x) = r(t)\dot{x} + x\frac{dr(t)}{dt}. \tag{2}$$

Hence by comparing (1) and (2), it seems that we need to find $r(t)$ satisfies

$$\frac{dr(t)}{dt} = -r(t)a(t),$$

which is exactly **Form 3**. Specifically, we can use

$$r(t) = e^{\int^t -a(s)ds}$$

as the integral factor. Then (1) becomes

$$r(t)\dot{x} - r(t)a(t)x = \frac{d}{dt}(r(t)x) = r(t)b(t) \implies \int \frac{d}{dt}(r(t)x)dt = r(t)x = \int^t r(s)b(s)ds + k.$$

Therefore,

$$x(t) = [r(t)]^{-1} \left(\int^t r(s)b(s)ds + k \right),$$

which is exactly the solution for **Form 4**.

Example 3

Assume that the motion of capital, $\dot{k}$, satisfies the following equation

$$\dot{k}(t) = -c(t) + 0.03k(t),$$

where $c(t)$ is consumption.

- (a) Find the general solution to this ODE.
- (b) Let $k(0) = 100$. Find the particular solution to this ODE.

This question is Form 4, with $a(t) = 0.03$ (a constant) and $b(t) = -c(t)$ (a function of time).

So, we have the general solution

$$\begin{aligned} k(t) &= \left[m + \int^t b(s)e^{-\int^s a(u)du}ds \right] e^{\int^t a(s)ds} \\ &= \left[m - \int^t c(s)e^{-\int^s 0.03du}ds \right] e^{\int^t 0.03ds} \\ &= \left[m - \int^t c(s)e^{-0.03s}ds \right] e^{0.03t} \end{aligned}$$

where m is some constant.

When $t = 0$, $k(0) = 100$ and we can substitute $t = 0$ into the general solution to get $m = 100$, which is the particular solution:

$$k(t) = \left[100 - \int_0^t c(s)e^{-0.03s} ds \right] e^{0.03t}$$

SEPARABLE ODES

In addition to linear ODEs, the separable ODEs also have explicit solutions.

Definition 2

A differential equation is **separable** if there exists g and h such that

$$F(t, x) = g(x) \cdot h(t).$$

To see how we solve the differential equation for separable ODEs, first write

$$\frac{dx}{dt} = g(x) \cdot h(t),$$

and then organize the terms,

$$\frac{1}{g(x)} \cdot \frac{dx}{dt} = h(t).$$

Then take the integral with respect to t ,

$$\int \frac{1}{g(x)} \cdot \frac{dx}{dt} dt = \int h(t) dt,$$

where we can apply change of variables by making $u = x(t)$ and $du(= dx) = \frac{dx}{dt} dt$ (chain rule). Hence

$$\int \frac{1}{g(x)} \cdot \frac{dx}{dt} dt = \int \frac{1}{g(x)} dx = \int h(t) dt.$$

Example 4

Let $u : \mathbb{R} \rightarrow \mathbb{R}$ be a utility function mapping a wealth level to some utility level. The **Arrow-Pratt measure of relative risk aversion** at x is defined as

$$r(x) = -\frac{u''(x)x}{u'(x)}.$$

Find the utility functions such that the relative risk aversion is a constant.

We write $u'(x) = v(x)$. Then we can solve the differential equation

$$r = -\frac{v'(x)x}{v(x)} \text{ or } \frac{dv}{dx} = -\frac{rv(x)}{x}.$$

Since the RHS is separable, we can rewrite

$$\frac{dv}{v(x)} = -r \frac{dx}{x} \text{ or } \int \frac{dv}{v(x)} = -r \int \frac{dx}{x}.$$

Hence we can solve

$$\log v(x) + C_v = -b(\log x + C_x).$$

Let $C = -bC_x - C_v$. Then

$$v(x) = x^{-b} \cdot e^C.$$

Let $k_1 = e^C$. Then for some k_2 ,

$$u(x) = \int v(x)dx = \int k_1 x^{-b} = \begin{cases} k_2 + k_1 \log x & \text{if } b = 1, \\ k_2 + \frac{k_1}{1-b} x^{1-b} & \text{if } b \neq 1. \end{cases}$$

SYSTEM OF DIFFERENTIAL EQUATIONS

Consider a scenario of two species of animals: rabbits ($x(t)$) and moles ($y(t)$). When the specie has a larger population, the population growth rate slows down. In addition, as rabbits and moles are competing the food on the grass on the shore, the population of the competing species also deteriorates the growth rate.

We can model the scenario as the follows.

$$\begin{cases} \frac{\dot{x}}{x} = a - bx - cy \\ \frac{\dot{y}}{y} = d - ey - fx \end{cases}.$$

In these models, we have more than one correlated differential equations. This is called the **system of differential equations**. In the matrix form, we can write the system of (first order ordinary) differential equations as $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}, t)$.

Once again, the system of differential equations cannot always be solved analytically. Hence we introduce a special class of differential equation system: **linear system of ODEs**.

$$\begin{cases} \dot{x}_1 = a_{11}x_1 + \cdots + a_{1n}x_n \\ \vdots \\ \dot{x}_n = a_{n1}x_1 + \cdots + a_{nn}x_n \end{cases} \implies \begin{bmatrix} \dot{x}_1 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}.$$

We can also rewrite the linear sequence with the matrix form: $\dot{\mathbf{x}} = A\mathbf{x}$. It is called linear because $\mathbf{F}(\mathbf{x}, t) = A\mathbf{x}$ is a linear transformation.

When A is diagonal, that is, $a_{ij} = 0$ for all $i \neq j$, then the solution goes back to the single variable linear ODE. Specifically,

$$x_i(t) = c_i e^{a_{ii}t}$$

for each $i = 1, \dots, n$.

When A is not diagonal, we can try to *diagonalize* the matrix with eigenvalues and eigenvectors. Suppose that A has n distinct eigenvalues $r_1, \dots, r_n$ and the corresponding eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$. By definition of eigenvalues and eigenvectors,

$$A\mathbf{v}_i = r_i\mathbf{v}_i.$$

Let $P = [\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_n]$ be the matrix collects all eigenvectors, then

$$AP = PD, \quad \text{where } D = \begin{bmatrix} r_1 & 0 & \cdots & 0 \\ 0 & r_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & r_n \end{bmatrix}.$$

Since the eigenvalues are all different, the eigenvectors are independent to each other. There-

fore P is invertible, and

$$P^{-1}AP = D.$$

Now take $\mathbf{y} = P^{-1}\mathbf{x}$, then $\mathbf{x} = P\mathbf{y}$. Since derivatives are linear operators,

$$\begin{aligned}\dot{\mathbf{y}} &= P^{-1}\dot{\mathbf{x}} \\ &= P^{-1}A\mathbf{x} && \text{(By the setting } \dot{\mathbf{x}} = A\mathbf{x}\text{)} \\ &= P^{-1}AP\mathbf{y} = D\mathbf{y}.\end{aligned}$$

Since D is diagonal, the solution goes back to the “easy” case:

$$\dot{y}_i = r_i y_i \Rightarrow y_i(t) = c_i e^{r_i t}.$$

Note that $\mathbf{x} = P\mathbf{y}$. Hence

$$\begin{aligned}\mathbf{x}(t) &= P\mathbf{y}(t) \\ &= [\mathbf{v}_1 \ \dots \ \mathbf{v}_n] \begin{bmatrix} c_1 e^{r_1 t} \\ \vdots \\ c_n e^{r_n t} \end{bmatrix} \\ &= c_1 e^{r_1 t} \mathbf{v}_1 + \dots + c_n e^{r_n t} \mathbf{v}_n.\end{aligned}$$

Be very careful that this applies to only the cases with distinct eigenvalues.

Example 5

Solve the following system of differential equations.

$$\begin{aligned}\dot{x} &= 2x + y \\ \dot{y} &= -12x - 5y.\end{aligned}$$

We rewrite the system of differential equations as

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -12 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

Suppose the eigenvalue is r . Then for eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$,

$$\begin{bmatrix} 2 & 1 \\ -12 & -5 \end{bmatrix} \mathbf{v}_i = r \mathbf{I} \mathbf{v}_i \Rightarrow \begin{bmatrix} 2-r & 1 \\ -12 & -5-r \end{bmatrix} \mathbf{v}_i = \mathbf{0}.$$

Thus

$$\det \left(\begin{bmatrix} 2-r & 1 \\ -12 & -5-r \end{bmatrix} \right) = 0 \Rightarrow r = -1 \text{ or } -2$$

and the corresponding eigenvectors are $(1, -3)$ and $(3, -2)$. Therefore, the solution is:

$$(x(t), y(t)) = (c_1 e^{-t} + 3c_2 e^{-2t}, -3c_1 e^{-t} - 2c_2 e^{-2t}).$$

EQUILIBRIUM OF DIFFERENTIAL EQUATIONS

There are some points that won't change even though they are governed by the differential equations. Take the rabbits and moles population as an example:

$$\begin{cases} \dot{x} = x(a - bx - cy) \\ \dot{y} = y(d - ey - fx) \end{cases}.$$

When $(x, y) = (0, 0)$, $(a/b, 0)$, or $(0, d/e)$, the population of the two species does not change over time. In other words, $\dot{x} = \dot{y} = 0$. These points are called **steady states**, or in a more econ way, **equilibrium**.

Definition 3

Let

$$\dot{x}_1 = f_1(x_1, \dots, x_n)$$

$$\vdots = \quad \vdots$$

$$\dot{x}_n = f_n(x_1, \dots, x_n)$$

be the system of differential equations. A point $(x_1^*, \dots, x_n^*)$ is a **steady state**, a **stationary solution**, a **rest point**, or an **equilibrium** if

$$f_i(x_1^*, \dots, x_n^*) = 0$$

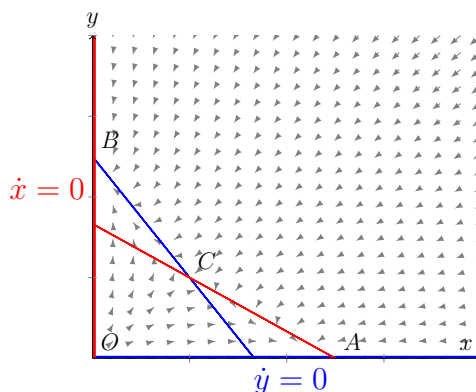
for each $i = 1, \dots, n$.

We can also rewrite the system of differential equations in vector form, $\dot{\mathbf{x}} = F(\mathbf{x})$. Then $\mathbf{x}^*$ is a steady state if $F(\mathbf{x}^*) = 0$.

We not only care about where the steady state is, but we also care about how “stable” it is. Take the rabbit vs moles example again. We can plot the graphs that $\dot{x} = 0$ and $\dot{y} = 0$. The intersections are the steady states.

$$\dot{x} = x(a - bx - cy) = 0 \quad \Leftrightarrow \quad x = 0 \text{ or } a - bx - cy = 0.$$

$$\dot{y} = y(d - ex - fy) = 0 \quad \Leftrightarrow \quad y = 0 \text{ or } d - ex - fy = 0.$$



The arrows in the diagram represent $\dot{x}$ and $\dot{y}$. They plot the “tendency” or the “dynamics” of the states, x and y . We focus on the $\dot{x}$ in the middle crossing the diagram first. For the region to the left of $\dot{x}$, the arrows are pointing to the right, which represents that $\dot{x} > 0$, or the dynamic at that point is *moving to the right*. Similarly, for the region to the right of $\dot{x}$, the arrows are pointing to the left, which represents that $\dot{x} < 0$. For the points on $\dot{x} = 0$, there is no movement toward left or right.

The interpretation for $\dot{y}$ is similar, except that $\dot{y}$ divides the plain into the “moving-up” and “moving-down” parts. We can then separate the plain into four regions. In the top-right region, the dynamics at the points are going bottom-left; in the bottom-left region, the dynamics at the points are going top-right.

We give a definition of stability.

Definition 4

Let $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}, t)$ be a system of differential equations, and $\phi(t)$ be a solution. We say a steady state $\mathbf{x}^*$ is stable if there exists $\varepsilon > 0$ such that for every solution $\phi(t)$,

$$\|\mathbf{x}^* - \phi(0)\| < \varepsilon \implies \lim_{t \rightarrow \infty} \phi(t) = \mathbf{x}^*.$$

From the diagram, you can find the four steady states. We first look at the one at the origin, O . The arrows around it are all moving outward, which means that when you move out a little bit from O , the dynamic will guide you away from O . Hence, O is **not stable**.

Contrarily, the arrows around A and B are leading the dynamic *back* to themselves. Hence we say A and B are **stable**.

C is somewhat tricky—if you move away to the top-right or bottom-left regions of C , it is guiding you back to C . However, if you move away to other parts, it can lead you away from C . In such case, we still say C is not stable; but there exists a solution which is stable.

There are conditions determining the stability of differential equations, but we will end our discussion about stability of differential equations here. We will study more ODE in macroeconomics.

Example 6 (Simplified Harrod-Domar model (Harrod 1939; Domar 1946))

Suppose the demand of a good is:

$$Q_d = \alpha - \beta P \quad (\alpha, \beta > 0)$$

and the supply of a good is:

$$Q_s = -\gamma + \delta P \quad (\gamma, \delta > 0)$$

Suppose $\dot{P} = j(Q_d - Q_s)$, where $j > 0$.

- What is the economic meaning of $(Q_d - Q_s)$? What does $\dot{P}$ mean in economic terms?
- Find the equilibrium price P^* .
- Solve the price function, $P(t)$. Is the equilibrium price P^* stable?

- The difference $(Q_d - Q_s)$ represents the *excess demand* or *shortage* in the market. If $Q_d > Q_s$, there is a shortage, indicating that demand exceeds supply. If $Q_d < Q_s$, there is excess supply or surplus.
 $\dot{P}$ represents the change of price over time. The parameter $j > 0$ determines how quickly the price adjusts in response to excess demand or supply: When $Q_d > Q_s$ (excess demand), the price will rise. When $Q_d < Q_s$ (excess supply), the price will fall.
- The equilibrium is reached when $\dot{P} = 0$, which means $Q_d = Q_s$. Therefore, $P^* = \frac{\alpha + \gamma}{\beta + \delta}$.
- This is the Form 2 ($\dot{x} = ax + b$). By applying the formula of the general solution;

$$P(t) = \frac{\alpha + \gamma}{\beta + \delta} + Ce^{-j(\beta + \delta)t}, \quad \text{where } C \text{ is some constant.}$$

Note that $P(t) \rightarrow \frac{\alpha + \gamma}{\beta + \delta}$ when $t \rightarrow \infty$ regardless of the value of C (i.e., regardless of the initial condition) because $j, \beta, \delta > 0$. So $P^* = \frac{\alpha + \gamma}{\beta + \delta}$ is stable.